

Caleb Bellsmith

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TECHNICAL SKILLS

Languages: C++, HTML, L^AT_EX, Visual Basic

Software/Tools: SolidWorks, Fusion360, AutoCAD, VSCode, CLion, Bamboo, Arduino IDE

Fabrication: CNC Machining, 3D Printing, Soldering, Manual Prototyping

PROJECTS

Hand Exoskeleton Rehabilitation

SolidWorks, ESP32, EMG Sensors

Independent Project

2025 – Present

- Created a mechanical orthotic exoskeleton aimed at rehabilitating hand and wrist injuries; optimized design for minimal weight and tool-less adjustability using SolidWorks modeling.
- Developing a motorized version using EMG sensors to detect neuromuscular signals for automated hand control.

Tilting Maze Platform

ESP32, C++, Computer Vision

Independent Project

Dec 2025 – Present

- Developed a modular, servo-powered tilting platform utilizing an ESP32 microcontroller and custom-built circuitry.
- Implementing a webcam-based computer vision system to enable autonomous maze-solving and real-time pathfinding.

Custom Cedar Strip Canoe

CAD, CNC Machining

Personal Project

2024 – Present

- Computer-modeled and fabricated a custom cedar strip canoe from scratch.
- Integrated CNC-machined internal molds with traditional woodworking to achieve high structural efficiency.

Adjustable Finger Prosthetic

User Testing, CAD, Iterative Design

Independent Project

2025 – Present

- Developed a functional finger prosthetic for a patient with scleroderma.
- Conducted repeated user-testing sessions to iterate on comfort and durability with interchangeable tips.

EXPERIENCE

True North Bio-competition

Electrical Designer

2025 – Present

- Creating the electrical control framework for a lower-limb exoskeleton, utilizing EMG sensors to facilitate intuitive, signal-based movement for assistive applications.

BioTEC Conference

Sponsorship Lead

Dec 2025 – Present

- Securing partnerships and corporate outreach for Canada's largest student-run biotechnology conference.

Lifeguard & Swimming Instructor

Instructor, NLS, CPR Certified

Town of Milton

June 2023 – 2025

- Selected for highly competitive outdoor pool placement (Top 4% of 250+ staff) based on performance.

Biomechatronics Design Team

Mechanical Designer

2025 – Present

- Developing mechanical components for assistive orthotic devices, focusing on structural integrity.

EDUCATION EXPERIENCE

University of Waterloo

B.A.Sc. in Biomedical Engineering

Waterloo, ON

Class of 2030

Honors: Governor General's Academic Medal, Sir Isaac Newton Physics Award, TDCH Valedictorian, HHCS Valedictorian.

WEEF Representative

Waterloo Engineering Endowment Foundation

Sept 2025 – Present

- Evaluated technical proposals for student project funding and represented class interests to faculty.

Personal Tutor

Mathematics and Physics

Sept 2023 – June 2025

- Instructed high school students in calculus, physics, biology, and chemistry, translating complex concepts.